

ONLINE TICKET BOOKING - WEB APPLICATION

ONLINE TICKET BOOKING-PS35

Problem Statement : ONLINE TICKET BOOKING-PS35

AIM :

The aim of the Online Ticket Booking Web Application is to develop a user-friendly and secure full-stack web application that enables users to browse events, check real-time ticket availability, select seats, book tickets, make online payments, and receive digital tickets conveniently.

The system also aims to provide administrators and event organizers with an efficient platform to manage events, venues, seats, ticket prices, bookings, and sales through a centralized dashboard.

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TEAM MEMBERS:

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We developed a online ticket booking upskill page, for Upskilling of technical skills.

SCOPE: The Online Ticket Booking Web Application provides a centralized platform for users to discover and book tickets for movies, concerts, sports, and other events. The system covers the complete booking process, from event selection to digital ticket generation.

1. **User Registration & Login:** Users can create accounts, log in securely, and manage their profiles.
2. **Event Browsing:** Users can view available movies, concerts, sports, and other events.
3. **Search & Filtering:** Users can search events based on category, location, date, and price.
4. **Event Details:** Users can view event descriptions, venue, date, time, ticket price, and availability.
5. **Seat Selection:** Users can view the seating layout and select available seats.
6. **Online Booking:** Users can book one or more tickets through the application.
7. **Online Payment:** The system can integrate secure payment gateways for completing transactions.
8. **Digital Ticket Generation:** After successful payment, the system generates an electronic ticket containing booking details and a unique booking ID/QR code.
9. **Booking History:** Users can view their previous and upcoming bookings.
10. **Admin/Organizer Management:** Administrators can add, update, or remove events, manage venues, ticket prices, seats, and bookings.
11. **Real-Time Availability:** Seat/ticket availability can be updated after successful bookings to reduce duplicate bookings.
12. **Reports & Analytics:** Organizers can monitor ticket sales, bookings, revenue, and event performance.

Project Modules:

ADMIN PAGE:
EVENT MANAGER:
USER REGISTRATION:

TECHNOLOGY STACK:

Scripting Languages: Hyper Text Markup Language -HTML
JavaScript + ECMA Script Features
CSS- Cascading Style Sheets

Frameworks : React.js + Node.js + Express.js + MongoDB.

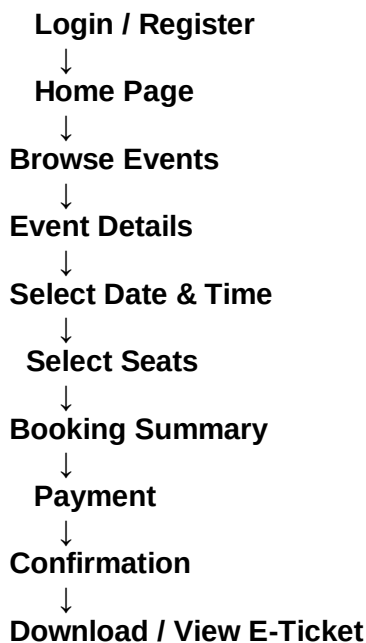
Design Thinking & Innovation Tool used:

Empathy Map, User Persona, Customer Journey Map (CJM), Problem Statement, HMW, Brainstorming, Prototyping, and User Testing.

Survey/Poll

Project Description:

a) Module Diagram / Prototype Diagram :



b) Implementation Module description:

Initial Level or Stage , we developed Responsive Web Page/Web app which is visible in all devices (Mobiles, Laptops or Larger Screens)

```
``html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>TicketBook - Online Ticket Booking</title>

<style>
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
    font-family: Arial, sans-serif;
}

body {
    background: #f5f7fb;
    color: #222;
}

/* NAVBAR */
.navbar {
    background: #111827;
    color: white;
    padding: 16px 6%;
    display: flex;
    align-items: center;
    justify-content: space-between;
}

.logo {
    font-size: 25px;
    font-weight: bold;
    color: #ff4757;
}

.navbar a {
    color: white;
    text-decoration: none;
    margin-left: 25px;
}

.navbar a:hover {
    color: #ff4757;
```

```
}
```

```
/* HERO */
```

```
.hero {  
  background: linear-gradient(135deg, #111827, #374151);  
  color: white;  
  text-align: center;  
  padding: 70px 20px;  
}
```

```
.hero h1 {  
  font-size: 42px;  
  margin-bottom: 15px;  
}
```

```
.hero p {  
  font-size: 18px;  
  margin-bottom: 30px;  
}
```

```
.search-box {  
  max-width: 650px;  
  margin: auto;  
  display: flex;  
  background: white;  
  border-radius: 8px;  
  overflow: hidden;  
}
```

```
.search-box input {  
  flex: 1;  
  padding: 16px;  
  border: none;  
  outline: none;  
  font-size: 16px;  
}
```

```
.search-box button {  
  padding: 16px 25px;  
  border: none;  
  background: #ff4757;  
  color: white;  
  cursor: pointer;  
}
```

```
/* EVENTS */
.container {
  width: 88%;
  max-width: 1200px;
  margin: 45px auto;
}

.section-title {
  text-align: center;
  margin-bottom: 30px;
}

.section-title h2 {
  font-size: 30px;
}

.events {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: 25px;
}

.card {
  background: white;
  border-radius: 12px;
  overflow: hidden;
  box-shadow: 0 5px 20px rgba(0,0,0,0.08);
  transition: 0.3s;
}

.card:hover {
  transform: translateY(-5px);
}

.card-image {
  height: 180px;
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 60px;
  background: #e5e7eb;
}

.card-content {
  padding: 20px;
```

```
}
```

```
.card-content h3 {  
  margin-bottom: 10px;  
}
```

```
.card-content p {  
  color: #666;  
  margin: 7px 0;  
}
```

```
.price {  
  color: #ff4757;  
  font-weight: bold;  
  font-size: 18px;  
}
```

```
.book-btn {  
  width: 100%;  
  padding: 12px;  
  margin-top: 15px;  
  border: none;  
  border-radius: 6px;  
  background: #ff4757;  
  color: white;  
  cursor: pointer;  
  font-size: 16px;  
}
```

```
.book-btn:hover {  
  background: #e63946;  
}
```

```
/* BOOKING */
```

```
.booking {  
  background: white;  
  padding: 30px;  
  border-radius: 12px;  
  box-shadow: 0 5px 20px rgba(0,0,0,0.08);  
  margin-top: 50px;  
}
```

```
.booking h2 {  
  margin-bottom: 25px;  
  text-align: center;
```

```
}
```

```
.form-group {  
  margin-bottom: 20px;  
}
```

```
.form-group label {  
  display: block;  
  margin-bottom: 8px;  
  font-weight: bold;  
}
```

```
.form-group input,  
.form-group select {  
  width: 100%;  
  padding: 13px;  
  border: 1px solid #ddd;  
  border-radius: 6px;  
}
```

```
/* SEATS */
```

```
.seats {  
  display: grid;  
  grid-template-columns: repeat(8, 1fr);  
  gap: 10px;  
  max-width: 500px;  
  margin: 20px auto;  
}
```

```
.seat {  
  padding: 10px;  
  text-align: center;  
  background: #e5e7eb;  
  border-radius: 5px;  
  cursor: pointer;  
}
```

```
.seat.selected {  
  background: #ff4757;  
  color: white;  
}
```

```
.seat.booked {  
  background: #555;  
  color: white;
```

```
    cursor: not-allowed;
}
```

```
.screen {
    text-align: center;
    background: #111827;
    color: white;
    padding: 10px;
    max-width: 500px;
    margin: 20px auto;
    border-radius: 5px;
}
```

```
/* CONFIRMATION */
```

```
.confirmation {
    display: none;
    background: #ecfdf5;
    border: 2px solid #10b981;
    padding: 25px;
    margin-top: 30px;
    border-radius: 10px;
}
```

```
.confirmation h3 {
    color: #059669;
    margin-bottom: 15px;
}
```

```
.ticket {
    background: white;
    padding: 20px;
    border-radius: 8px;
    margin-top: 15px;
}
```

```
.footer {
    background: #111827;
    color: white;
    text-align: center;
    padding: 25px;
    margin-top: 60px;
}
```

```
/* RESPONSIVE */
```

```
@media (max-width: 900px) {
```

```
.events {
  grid-template-columns: repeat(2, 1fr);
}
}
```

```
@media (max-width: 600px) {
```

```
.navbar {
  flex-direction: column;
  gap: 15px;
}
```

```
.navbar a {
  margin: 0 8px;
}
```

```
.hero h1 {
  font-size: 30px;
}
```

```
.search-box {
  flex-direction: column;
}
```

```
.search-box button {
  width: 100%;
}
```

```
.events {
  grid-template-columns: 1fr;
}
```

```
.seats {
  grid-template-columns: repeat(4, 1fr);
}
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<!-- NAVIGATION -->
```

```
<nav class="navbar">
```

```
  <div class="logo"> TicketBook</div>
```

```

<div>
  <a href="#home">Home</a>
  <a href="#events">Events</a>
  <a href="#booking">Booking</a>
</div>
</nav>

<!-- HERO -->
<section class="hero" id="home">

  <h1>Book Your Tickets Easily</h1>

  <p>
    Movies • Concerts • Sports • Events
  </p>

  <div class="search-box">
    <input
      type="text"
      id="searchInput"
      placeholder="Search movies, concerts, sports..."
    >

    <button onclick="searchEvents()">🔍 Search</button>
  </div>

</section>

<!-- EVENTS -->
<section class="container" id="events">

  <div class="section-title">
    <h2>Popular Events</h2>
    <p>Choose your favorite event and book your tickets</p>
  </div>

  <div class="events" id="eventList">

    <div class="card" data-name="Avengers Movie">
      <div class="card-image">🎬</div>

      <div class="card-content">
        <h3>Avengers Movie</h3>
        <p>📍 PVR Cinemas</p>
        <p>📅 15 October 2026</p>
      </div>
    </div>
  </div>

```

```
<p class="price">₹250</p>

<button
  class="book-btn"
  onclick="selectEvent('Avengers Movie', 250)"
>
  Book Now
</button>
</div>
</div>

<div class="card" data-name="Music Concert">
  <div class="card-image">🎵</div>

  <div class="card-content">
    <h3>Music Concert</h3>
    <p>📍 Vijayawada Stadium</p>
    <p>📅 20 October 2026</p>
    <p class="price">₹999</p>

    <button
      class="book-btn"
      onclick="selectEvent('Music Concert', 999)"
    >
      Book Now
    </button>
  </div>
</div>

<div class="card" data-name="Cricket Match">
  <div class="card-image">🏏</div>

  <div class="card-content">
    <h3>Cricket Match</h3>
    <p>📍 ACA Stadium</p>
    <p>📅 25 October 2026</p>
    <p class="price">₹500</p>

    <button
      class="book-btn"
      onclick="selectEvent('Cricket Match', 500)"
    >
      Book Now
    </button>
  </div>
</div>
```

</div>

</div>

<!-- BOOKING -->

<div class="booking" id="booking">

<h2>🎫 Book Your Ticket</h2>

<div class="form-group">

<label>Selected Event</label>

<input

type="text"

id="selectedEvent"

placeholder="Select an event above"

readonly

>

</div>

<div class="form-group">

<label>Select Date</label>

<input

type="date"

id="bookingDate"

>

</div>

<div class="form-group">

<label>Your Name</label>

<input

type="text"

id="userName"

placeholder="Enter your name"

>

</div>

<div class="form-group">

<label>Email</label>

<input

type="email"

id="userEmail"

```

        placeholder="Enter your email"
    >
</div>

<div class="screen">
    SCREEN
</div>

<h3 style="text-align:center;">Select Your Seats</h3>

<div class="seats">

    <div class="seat" onclick="selectSeat(this)">A1</div>
    <div class="seat" onclick="selectSeat(this)">A2</div>
    <div class="seat booked">A3</div>
    <div class="seat" onclick="selectSeat(this)">A4</div>
    <div class="seat" onclick="selectSeat(this)">A5</div>
    <div class="seat" onclick="selectSeat(this)">A6</div>
    <div class="seat booked">A7</div>
    <div class="seat" onclick="selectSeat(this)">A8</div>

    <div class="seat" onclick="selectSeat(this)">B1</div>
    <div class="seat" onclick="selectSeat(this)">B2</div>
    <div class="seat" onclick="selectSeat(this)">B3</div>
    <div class="seat booked">B4</div>
    <div class="seat" onclick="selectSeat(this)">B5</div>
    <div class="seat" onclick="selectSeat(this)">B6</div>
    <div class="seat" onclick="selectSeat(this)">B7</div>
    <div class="seat" onclick="selectSeat(this)">B8</div>

    <div class="seat" onclick="selectSeat(this)">C1</div>
    <div class="seat" onclick="selectSeat(this)">C2</div>
    <div class="seat" onclick="selectSeat(this)">C3</div>
    <div class="seat" onclick="selectSeat(this)">C4</div>
    <div class="seat booked">C5</div>
    <div class="seat" onclick="selectSeat(this)">C6</div>
    <div class="seat" onclick="selectSeat(this)">C7</div>
    <div class="seat" onclick="selectSeat(this)">C8</div>

</div>

<p style="text-align:center; margin:20px;">
    Selected Seats:
    <strong id="seatCount">0</strong>
</p>

```

```
<p style="text-align:center; margin:20px;">
  Total Amount:
  <strong>₹<span id="totalPrice">0</span></strong>
</p>
```

```
<button class="book-btn" onclick="confirmBooking()">
   Confirm Booking
</button>
```

```
<!-- CONFIRMATION -->
<div class="confirmation" id="confirmation">
```

```
  <h3> Booking Confirmed!</h3>
```

```
  <div class="ticket">
```

```
    <p><strong>Booking ID:</strong>
      <span id="bookingId"></span>
    </p>
```

```
    <p><strong>Name:</strong>
      <span id="ticketName"></span>
    </p>
```

```
    <p><strong>Event:</strong>
      <span id="ticketEvent"></span>
    </p>
```

```
    <p><strong>Date:</strong>
      <span id="ticketDate"></span>
    </p>
```

```
    <p><strong>Seats:</strong>
      <span id="ticketSeats"></span>
    </p>
```

```
    <p><strong>Total:</strong>
      ₹<span id="ticketPrice"></span>
    </p>
```

```
  <br>
```

```
<p style="text-align:center;">
   Thank you for booking with TicketBook!
```

</p>

</div>

</div>

</div>

</section>

<footer class="footer">

<p>© 2026 TicketBook | Online Ticket Booking System</p>

</footer>

<script>

let selectedEventName = "";

let ticketPrice = 0;

let selectedSeats = [];

/* SELECT EVENT */

function selectEvent(name, price) {

selectedEventName = name;

ticketPrice = price;

document.getElementById("selectedEvent").value = name;

document

.getElementById("booking")

.scrollIntoView({ behavior: "smooth" });

updateTotal();

}

/* SELECT SEAT */

function selectSeat(seat) {

if (seat.classList.contains("booked")) {

return;

}

const seatNumber = seat.innerText;

```

if (seat.classList.contains("selected")) {

    seat.classList.remove("selected");

    selectedSeats =
        selectedSeats.filter(
            s => s !== seatNumber
        );

} else {

    seat.classList.add("selected");

    selectedSeats.push(seatNumber);
}

updateTotal();
}

/* UPDATE PRICE */

function updateTotal() {

    document.getElementById("seatCount").innerText =
        selectedSeats.length;

    document.getElementById("totalPrice").innerText =
        selectedSeats.length * ticketPrice;
}

/* SEARCH */

function searchEvents() {

    const search =
        document
        .getElementById("searchInput")
        .value
        .toLowerCase();

    const cards =
        document.querySelectorAll(".card");

    cards.forEach(card => {

```

```
const name =
  card
  .getAttribute("data-name")
  .toLowerCase();

if (name.includes(search)) {

  card.style.display = "block";

} else {

  card.style.display = "none";
}

});
}

/* CONFIRM BOOKING */

function confirmBooking() {

  const name =
    document.getElementById("userName").value;

  const email =
    document.getElementById("userEmail").value;

  const date =
    document.getElementById("bookingDate").value;

  if (selectedEventName === "") {

    alert("Please select an event.");

    return;
  }

  if (name === "" || email === "") {

    alert("Please enter your name and email.");

    return;
  }
}
```

```
if (date === "") {

    alert("Please select a date.");

    return;
}

if (selectedSeats.length === 0) {

    alert("Please select at least one seat.");

    return;
}

const bookingNumber =
    "TB" + Math.floor(100000 + Math.random() * 900000);

document.getElementById("bookingId").innerText =
    bookingNumber;

document.getElementById("ticketName").innerText =
    name;

document.getElementById("ticketEvent").innerText =
    selectedEventName;

document.getElementById("ticketDate").innerText =
    date;

document.getElementById("ticketSeats").innerText =
    selectedSeats.join(", ");

document.getElementById("ticketPrice").innerText =
    selectedSeats.length * ticketPrice;

document.getElementById("confirmation").style.display =
    "block";

document
    .getElementById("confirmation")
    .scrollIntoView({ behavior: "smooth" });

}

</script>
```

</body>

</html>

```

Youtube Demonstration of Project is available in public url : [link](#)

Github Repository of Project is available as public repository : [link](#)